

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						
Q10						

Code of Conduct:

I, G Madhulatha(192311295), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G Madhulatha

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formalisms [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

Q1. Define Data Mining and list any two basic data mining tasks. Explain the KDD process and role of preprocessing. [L2, L4]

ANSWER:

Data Mining is the process of discovering useful patterns and knowledge from large datasets.

Two basic tasks:

1. Classification – Assigns data into predefined classes.
2. Clustering – Groups similar data without predefined classes.

KDD Process:

1. Data Selection – Select relevant data.
2. Data Cleaning – Remove errors, missing values, and noise.
3. Data Integration – Combine data from different sources.
4. Data Transformation – Convert data into a suitable format.
5. Data Mining – Discover useful patterns.
6. Pattern Evaluation – Identify interesting patterns.
7. Knowledge Presentation – Present the discovered knowledge.

Role of Preprocessing:

Preprocessing improves data quality by handling missing values, noise, outliers, and inconsistencies. It improves the accuracy and efficiency of data mining.

Q2. What is data cleaning? Mention two data quality issues. Explain methods to handle missing and noisy data. [L1, L2, L3]

ANSWER:

Data cleaning is the process of detecting and correcting inaccurate, incomplete, duplicate, and inconsistent data.

Two data quality issues:

1. Missing values.
2. Noisy or incorrect values.

Methods for Missing Values:

1. Mean/Median – Replace missing numerical values.
2. Mode – Replace missing categorical values.
3. k-NN – Use similar records to estimate the missing value.
4. Delete the record if only a few values are missing.

Methods for Noisy Data:

1. Binning – Smooth values using bin mean or median.
2. Regression – Estimate correct values.
3. Outlier detection – Identify extreme values using Z-score or IQR.
4. Clustering – Detect unusual data points.

Example:

Ages = 20, 22, 25, 250

Here, 250 is an extreme/noisy value.

=====

Q3. Define data integration and list two challenges. Explain how redundancy and inconsistency are handled. [L2, L4]

ANSWER:

Data integration is the process of combining data from multiple sources into one consistent dataset.

Two challenges:

1. Schema differences – Different column names or structures.
2. Data inconsistency – Different formats or values for the same information.

Handling Redundancy:

1. Remove duplicate records.
2. Remove unnecessary duplicate attributes.
3. Use correlation to identify redundant features.

Handling Inconsistency:

1. Standardize formats.
2. Convert units into a common format.
3. Standardize categorical values.

Example:

M, Male, male → Male

=====

Q4. What is covariance analysis? List the two covariance formulas. [L2, L3]

ANSWER:

Covariance analysis determines the direction of the relationship between two variables.

Positive covariance → Variables move in the same direction.

Negative covariance → Variables move in opposite directions.

Zero covariance → Little or no linear relationship.

Population Covariance:

$$\text{Cov}(X,Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] / N$$

Sample Covariance:

$$\text{Cov}(X,Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] / (N - 1)$$

=====

**Q5. Stocks A and B have values:
(2,5), (3,8), (5,10), (4,11), (6,14).**

Will their prices rise or fall together or independently?

ANSWER:

Mean of A:

$$\bar{A} = (2+3+5+4+6)/5 = 4$$

Mean of B:

$$\bar{B} = (5+8+10+11+14)/5 = 9.6$$

Covariance:

$$\text{Cov}(A,B) = \Sigma[(A_i - \bar{A})(B_i - \bar{B})] / 5$$

$$= 20 / 5$$

$$= +4$$

Since covariance is positive, the two stocks tend to move in the same direction.

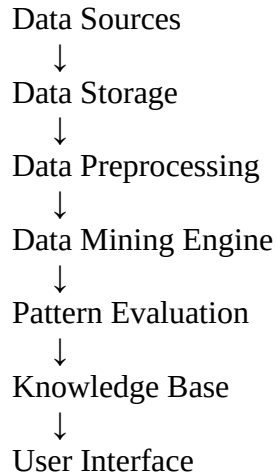
Therefore, their prices are likely to rise together or fall together rather than move independently.

=====

Q6. Explain the architecture of a typical data mining system. [L2, L4]

ANSWER:

Basic Architecture:



Components:

1. Data Sources – Provide raw data.
2. Data Storage – Stores databases and data warehouses.
3. Preprocessing – Cleans, integrates, and transforms data.
4. Data Mining Engine – Performs classification, clustering, association, etc.
5. Pattern Evaluation – Identifies useful patterns.
6. Knowledge Base – Stores domain knowledge and rules.
7. User Interface – Displays results using reports and graphs.

Interaction:

Raw data is collected, cleaned, transformed, mined for patterns, evaluated, and finally presented as useful knowledge.

=====

Q7. Find covariance for:

$X = \{4, 5, 6, 7, 9\}$

$Y = \{8, 3, 7, 5, 6\}$

ANSWER:

Mean of X:

$$\bar{X} = 31/5 = 6.2$$

Mean of Y:

$$\bar{Y} = 29/5 = 5.8$$

Sum of products:

$$\Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] = -1.80$$

Population covariance:

$$\text{Cov}(X, Y) = -1.80/5$$

$$\text{Cov}(X, Y) = -0.36$$

FINAL ANSWER:

$$\text{Covariance} = -0.36$$

Therefore, X and Y have a negative relationship.

For sample covariance:

$$\text{Cov}(X, Y) = -1.80/4 = -0.45$$

=====

Q8. Describe the major steps in data preprocessing. Propose a systematic preprocessing pipeline. [L3, L4, L5]

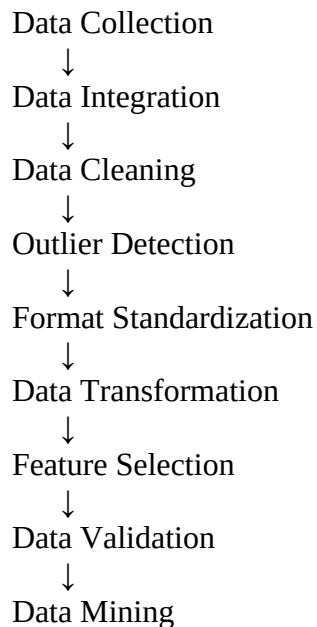
ANSWER:

Major Steps:

1. Data Cleaning – Handle missing, noisy, and duplicate data.
2. Data Integration – Combine data from multiple sources.
3. Data Transformation – Apply normalization or standardization.
4. Data Reduction – Remove irrelevant or redundant features.

5. Data Discretization – Convert continuous data into categories.

Suggested Pipeline:



Justification:

Cleaning should be performed before analysis to avoid errors. Outliers are treated after cleaning, followed by standardization and transformation. Feature selection removes unnecessary information before mining.

=====

Q9. Perform equal-frequency binning on the following age data:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

ANSWER:

Number of observations = 27

Number of bins = 3

Records per bin = $27/3 = 9$

Bin 1:

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2:

22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3:

35, 35, 35, 36, 40, 45, 46, 52, 70

Using bin mean for smoothing:

Bin 1 Mean = $162/9 = 18$

Bin 2 Mean = $253/9 = 28.11$

Bin 3 Mean = $394/9 = 43.78$

Smoothed Data:

Bin 1 $\rightarrow$ 18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2 $\rightarrow$ 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11

Bin 3 $\rightarrow$ 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78, 43.78

=====

Q10. Explain data discretization. Compare equal-width and equal-frequency binning. [L3, L4, L5]

ANSWER:

Data discretization is the process of converting continuous numerical values into a smaller number of intervals or categories.

Importance:

1. Reduces data complexity.
2. Makes data easier to understand.
3. Reduces noise.
4. Helps classification and rule-based mining.
5. Improves visualization.

Equal-Width Binning:

The data range is divided into intervals of equal numerical width.

Formula:

$$\text{Bin Width} = (\text{Maximum} - \text{Minimum}) / \text{Number of Bins}$$

Example:

Data = 10, 20, 30, 40, 50, 60

For 3 bins:

$$\text{Bin Width} = (60-10)/3 = 16.67$$

Advantages:

1. Simple to implement.
2. Equal numerical ranges.
3. Suitable for uniformly distributed data.

Disadvantage:

Bins may contain different numbers of records.

Equal-Frequency Binning:

The data is divided so that each bin contains approximately the same number of records.

Example:

Bin 1 → 10, 20

Bin 2 → 30, 40

Bin 3 → 50, 60

Advantages:

1. Equal or nearly equal number of records.
2. Useful for skewed data.
3. Produces balanced groups.

Disadvantage:

Bin widths may be different.

Comparison:

Equal-Width:

Equal numerical range, unequal number of records.

Equal-Frequency:

Equal number of records, unequal numerical range.

CONCLUSION:

Discretization simplifies continuous data. Equal-width binning is suitable for uniformly distributed data, while equal-frequency binning is better for skewed data and balanced grouping.